Final Assignment – FinTech Course

The final assignment requires students to address a given problem starting from the assigned dataset.

The delivery must include a well-structured code implementation (in a Jupyter Notebook or Google Colab Notebook). The code should be clearly commented, or alternatively accompanied by a written report describing the methodology, implementation, and results.

In addition, students are required to prepare a PowerPoint presentation that frames the problem, describe the dataset, explains the approach taken, highlights the main results, and concludes with key insights and considerations.

Your assignment: Credit Card Customer (Dataset 1)

A manager at the bank is disturbed with more and more customers leaving their credit card services. They would really appreciate if one could predict for them who is gonna get churned so they can proactively go to the customer to provide them better services and turn customers' decisions in the opposite direction

This dataset consists of 10,000 customers mentioning their age, salary, marital_status, credit card limit, credit card category, etc. There are nearly 18 features.

The dataset

1. Client number. Unique identifier for the customer holding the account
2. Internal event (customer activity) variable - if the account is closed then 1 else 0
3. Demographic variable - Customer's Age in Years
4. Demographic variable - M=Male, F=Female
5. Demographic variable - Number of dependents
6. Demographic variable - Educational Qualification of the account holder (example: high school, college graduate, etc.)
7. Demographic variable - Married, Single, Divorced, Unknown
8. Demographic variable - Annual Income Category of the account holder (< $40K, $40K - 60K, $60K - $80K, $80K-$120K, > $120K, Unknown)
9. Product Variable - Type of Card (Blue, Silver, Gold, Platinum)
10. Period of relationship with bank
11. Total no. of products held by the customer
12. No. of months inactive in the last 12 months
13. No. of Contacts in the last 12 months
14. Credit Limit on the Credit Card
15. Total Revolving Balance on the Credit Card
16. Open to Buy Credit Line (Average of last 12 months)
17. Change in Transaction Amount (Q4 over Q1)
18. Total Transaction Amount (Last 12 months)
19. Total Transaction Count (Last 12 months)
20. Change in Transaction Count (Q4 over Q1)
21. Average Card Utilization Ratio

What to do:

- *apply Lasso regression to reduce the number of features, and apply a random forest classifier to the reduced dataset, choosing the optimal hyperparameters.*
- *compare the results in terms of prediction with a random forest classifier on the whole set of original features, choosing the optimal hyperparameters.*
- *compare the best model with a XGBoost classifier.*